

A Survey of Low-light Image Enhancement Algorithms based on Deep Learning

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Abstract—Low-light image enhancement (LLIE) represents a critical challenge in computer vision, focusing on improving image visibility while preserving fine details and natural color fidelity. While traditional approaches like histogram equalization and Retinex-based methods have been widely used, they often suffer from noise amplification and color distortion artifacts. Recent advances in deep learning have revolutionized this field, achieving unprecedented performance. However, the rapid proliferation and diverse nature of these deep learning techniques necessitate a comprehensive and systematic review to consolidate current knowledge, structure the evolving landscape, and identify key trends and challenges. This paper systematically reviews state-of-the-art deep learning methods for LLIE, including convolutional neural network (CNN)-based architectures, generative adversarial networks (GANs), and Transformer models. Furthermore, we comprehensively analyze benchmark datasets, standardized evaluation metrics, and persistent challenges in the field, while also outlining promising directions for future research.

Keywords—Low-light image enhancement, survey, Convolutional Neural Network, Generative Adversarial Networks, Deep Learning.

I. INTRODUCTION

Computer vision processing needs clear pictures to easily get information, as the devices price are going down, computer vision systems have been widely used in various scenarios, such as object detection, face recognition, image classification, and visual surveillance [1]. However, image acquisition is often affected by uncontrollable factors, particularly under insufficient lighting conditions such as indoors or at night, leading to various image defects (e.g., loss of detail, poor resolution) [2]. Therefore, improve the visual quality of images collected in low-light or dim conditions by algorithmic enhancement is necessary, as this can be automatic and has low cost.

Low-light image enhancement (LLIE) algorithms can be separated into two types: traditional algorithms and deep learning-based algorithms. Traditional algorithms (Histogram equalization (HE) [3] and Retinex theory-based methods [4]) offer advantages such as no need to train and fast speed and ease of deployment, but they lack reference to real-world illumination conditions, leading to issues like noise retention or amplification, artifacts, and color deviations.

Histogram equalization methods enhance image brightness and contrast by making the grayscale distribution more uniform, but this algorithm is prone to over-

enhancement and noise amplification. Retinex based approaches decompose a low-light image into reflectance and illumination components through certain priors or regularization, treating the estimated reflectance component as the enhanced result. While effective in improving detail and contrast in low-light images, these methods are limited by their handcrafted prior design capabilities, and excessive constraints in joint optimization often led to compromised adaptability and efficiency. Consequently, Retinex based methods frequently produce unnatural-looking images.

The shortcomings of traditional LLIE approaches spurred the development and ongoing refinement of various deep learning solutions, which have now become the standard. These deep learning techniques distinguish themselves from conventional methods by offering superior precision and enhanced stability.

In recent years, the rapid development of Deep Learning (DL) has opened up new avenues for addressing LLIE problems, achieving unprecedented performance. Models based on deep neural networks are capable of learning complex image priors and mapping relationships from large-scale data, thereby demonstrating significant potential in enhancing image brightness, suppressing noise, and improving detail. However, as research progresses, the number of DL-based LLIE algorithms has surged, with network architectures, learning strategies, and optimization objectives becoming increasingly diverse. This rapid technological iteration and the proliferation of methods make it challenging for researchers to comprehensively grasp the latest advancements, core technical frameworks, and future development trends in this field. Therefore, it is particularly urgent and important to conduct a systematic review, analysis, and summary of existing DL-based LLIE algorithms. This literature review aims to address this need, providing researchers with a clear and comprehensive reference framework.

This paper summarizes the latest research results in deep learning-based LLIE methods. It provides a review of representative deep learning methods, datasets, evaluation metrics, as well as qualitative and quantitative experimental results. This work serves as a valuable reference for beginners seeking to understand this research domain.

The core objective of this review is to provide a comprehensive framework for analyzing the latest advancements in the field of deep learning-based LLIE methods. Specifically, we aim to: (1) propose a clear and

insightful classification system to summarize and organize existing mainstream deep learning LLIE algorithms; (2) deeply analyze the core principles, network architectures, training strategies, and respective strengths and weaknesses of various representative algorithms; (3) systematically summarize and compare commonly used benchmark datasets and performance evaluation metrics, and discuss their applicability and limitations; (4) refine and discuss the key challenges and bottlenecks that persist in current research.

The main contributions of this paper include: (1) Comprehensive Summary of Latest Advancements: We systematically review and integrate the latest deep learning LLIE algorithms up to 2024. (2) Clear Classification and Comparison: Different methods are categorized based on learning approaches, and their effectiveness is comparatively analyzed. (3) Introduction to Datasets and Evaluation Methods: We organize datasets and evaluation methods used for low-light enhancement. (4) Clear Guidance for Future Directions: Based on summarizing existing challenges, we distill several promising future research directions and open problems, aiming to inspire and drive continuous innovation in this field.

II. LITERATURE REVIEW

The deep learning method uses neural networks to learn complex lighting adjustment relationships, which can automatically extract useful features and achieve more delicate enhancement effects. Compared with the traditional methods, the deep learning model can adapt to different lighting conditions and improve the generalization ability. In addition, the neural network can also combine the perception

loss function to optimize and enhance the visual quality of the image, making it more realistic and natural [5].

Initially, deep learning-based methods primarily utilized supervised learning approaches and achieved significant

success. Due to the difficulty in obtaining paired datasets of normal/low-light images, unsupervised learning methods have been developed. LLIE methods can be roughly categorized into four types based on training data: supervised learning, unsupervised learning, semi-supervised learning, and zero-shot learning. This section will provide a detailed introduction to representative deep learning-based LLIE methods, which are also presented in Table 1.

Among the methods in LLIE fields, Zero-DCE got the largest citations, because it came earlier, has good result, and it is unsupervised, lightweight, innovative, effective, and easy to use. LLIE is a complex problem, requiring fine-grained control, and zero-shot models lack sufficient external guidance, making it particularly challenging in adapting to all situations, result in few methods using Zero-shot models. Semi-supervised learning is less common in LLIE because the task of enhancement blurs the distinction between "labeled and unlabeled" data, making pseudo-label generation difficult. Additionally, low-light conditions inherently emphasize the complexity of restoration rather than being a simple classification problem. With paired data, clear objectives, stable training, and well-defined metrics, supervised methods are the most likely to succeed in low-light enhancement, which is why they are the most widely used. Unsupervised LLIE methods are numerous because single-image data is easily obtainable, the loss is controllable, the generalization ability is strong.

Table -1 LLIE methods based on deep learning

Year	Method	Learning	Network	Loss function	Citations	Publication
2020	Zero-DCE[6]	Zero-shot	Fully connected network	Spatial consistency loss; Color constancy loss; Illumination smoothness loss;	2033	CVPR
2020	DSLRL[7]	Supervised	Laplacian pyramid; U-Net like network	L2 loss; Color loss; Laplacian loss	253	IEEE
2021	DRBN[8]	Semi-supervised	Recursive network	SSIM loss; Perceptual loss; Adversarial loss	639	CVPR
2021	RUAS[9]	Zero-shot	Neural architecture search	Cooperative loss; Similar Loss; Total variation loss	883	CVPR
2022	SCI[10]	Unsupervised	Self-Calibrated Illumination	Fidelity loss; Smoothness loss	808	CVPR
2022	SNR-aware[11]	Supervised	SNR-guided attention	Charbonnier loss;	474	CVPR

Year	Method	Learning	Network	Loss function	Citations	Publication
				Perceptual loss		
2023	LLFormer[12]	Supervised	U-net; Transformer	L1 loss; Perceptual loss; SSIM loss; Color loss	287	CVPR
2023	PairLIE[13]	Unsupervised	Encoder- decoder networks	Projection Loss; Retinex Loss; Reflectance consistency loss	208	CVPR
2023	CUE[14]	Supervised	Masked autoencoder	smoothness loss; Noise prior loss	128	ICCV
2023	Dimma[15]	Semi-supervised	Mixture density network;	Mean squared error loss;	6	arxiv
2024	CDAN[16]	Supervised	Autoencoder	Perceptual loss; Mean squared error loss;	8	arxiv
2024	Cycle- Retinex[17]	Unsupervised	CycleGAN	Perceptual loss; Loss of CycleGAN; Loss of De-Net; Loss of RR-Net	25	IEEE

A. Supervised learning methods

Lim et al. [7] introduced a new method use the Laplacian pyramid (DSLRL) to restore global layout and local details in both image and feature spaces.

A new framework for LLIE was developed by Wang et al. [8], driven by the impressive feature representation observed in Masked Autoencoders. Their study investigates how incorporating customized, learnable priors can enhance transparency in the deep unfolding framework.

Hossein [16] developed the Convolutional Dense Attention-guided Network (CDAN), offering a new approach to enhance images captured in low-light conditions. CDAN employs an autoencoder-style framework that incorporates convolutional and dense layers, further supported by an attention mechanism and skip connections. This design enables effective data flow and optimized feature extraction.

Xu et al. [11] introduced a novel LLIE approach that combined SNR-aware transformers and convolutional networks to adaptively refine pixels using spatially adaptive operations. Their method employs an SNR prior to direct feature integration and develops an SNR-sensitive transformer with a custom self-attention mechanism, preventing interference from high-noise, low-SNR regions.

B. Zero-shot learning

Li et al. [6] proposed an algorithm called Zero-DCE (Zero-Reference Deep Curve Estimation) based on the concept of zero-shot learning, which treats LLIE as a problem of estimating an image's depth curve. They developed a deep high-order curve model that can automatically transform dark

images into enhanced ones, known as the light enhancement curve. By estimating the deep high-order curve of the input image, the network is guided to perform pixel-level adjustments on low-light images with the LE curve as the target.

The enhancement process of Zero-DCE is as follows: each pixel I is iteratively enhanced over T steps to become the enhanced image:

$$I_{t+1} = I_t + \alpha_t \cdot I_t \cdot (1 - I_t), \quad t = 0, 1, \dots, T - 1$$

This formula is a pixel-level self-enhancement curve function with excellent nonlinear adjustment capabilities. Among them,

I_0 : The original input image;

α_t : The curve parameters predicted by the network;

I_t : The enhanced result at the t -th step;

Zero-DCE++ [18] enhances model efficiency by further reducing the number of network layers and decreasing the coupling degree of contrast differences between pixels. This approach addresses the real-time processing challenges in low-light video enhancement without compromising the original Zero-DCE enhancement effects.

Ma et al. [9] introduced RUAS (Retinex-inspired Unfolding Architecture Search), a novel framework for designing efficient and lightweight LLIE networks customized for real-world applications. Inspired by Retinex theory, RUAS first formulates a model to represent the inherent underexposed properties of low-light images and

unfolds its optimization procedure to build the complete network architecture. Additionally, it employs a reference-free collaborative learning approach to identify optimal low-light enhancement structures within a constrained search space. This enables RUAS to achieve high-performance image enhancement with fast inference speeds and low computational demands.

C. Semi-Supervised learning methods

DRBN (Deep Recursive Band Network) [8] enhances the linear band representation of normal-light images by leveraging paired low/normal-light data. It then refines these bands via a trainable linear transformation guided by perceptual adversarial learning on unpaired data, ultimately producing an optimized band representation.

Separately, Kozlowski et al. [15] introduced DIMMA, an algorithm that employs semi-real images to build a dimming simulator replicating camera-specific underexposure. This module then generates artificially dimmed images from diverse datasets, which train a U-Net to recover proper illumination.

D. Unsupervised learning

Generative Adversarial Networks (GANs) [19] have proven effective for image-to-image translation tasks, including low-light enhancement. GANs leverage a generator network to produce enhanced images and a discriminator network to identify fake images from real normal-light images, leading to perceptually more convincing results. By adversarial training:

G aims to make $D(G(z)) \rightarrow 1$ (to make the discriminator believe the generated images are real).

D aims to make $D(x) \rightarrow 1$ and $D(G(z)) \rightarrow 0$ (to distinguish between real and fake images).

The formula of GAN is below:

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log (1 - D(G(z)))]$$

Among them, $p_{data}(x)$ is the distribution of real data. $p_z(z)$ is the input noise distribution for the generator (e.g., Gaussian or uniform distribution). $D(x)$ is the probability that the discriminator considers the image x to be real. $G(x)$ is the image generated by the generator.

MAL et al. [10] introduced Self-Calibrated Illumination Learning (SCI), with two main contributions: the use of a

hierarchical self-calibration module for weight-sharing illumination learning and unsupervised training, enabling the entire illumination enhancement model to achieve faster convergence; simultaneously, it ensures the correlation between low-light images and optimized images based on illumination components, suppressing overexposure.

Fu et al. [12] employed paired low-light images during training to maximize information extraction. While grounded in Retinex theory, their approach deliberately avoids conventional Retinex decomposition methods. Instead, a simple self-supervised mechanism is adopted to remove unreasonable features (usually some noise) and better realize the Retinex theory. Retinex (Retina + Cortex) was proposed by Edwin Land [28] to explain how the human visual system maintains color constancy under varying lighting conditions. Its core assumption is that an image $S(x, y)$ is composed of two parts: illumination $L(x, y)$ and reflectance $R(x, y)$. The formula of Retinex is below:

$$S(x, y) = R(x, y) \cdot L(x, y)$$

Wu et al. [17] proposed Cycle-Retinex, a novel LLIE method that generates adversarial networks through Retinex internal cycle consistency. This network is trained in an unsupervised manner. Cycle-Retinex organically combines Retinex theory and CycleGAN theory, dividing the low-light enhancement task into two separate subtasks: enhancing the illumination map and restoring the reflectance map.

III. DATASETS

Low-light datasets are a significant limiting factor affecting the robustness in various scenarios of deep learning-based methods. Existing LLIE datasets can be categorized into two types based on data format: RGB and RAW. RGB is the standard color image format processed by the ISP (Image Signal Processor), while RAW is the raw data directly obtained from the camera sensor.

The LOL dataset [21] contains 500 aligned image pairs of low-light and normal-light scenes captured indoors, with 485 pairs allocated for training and 15 for testing. In comparison, the SID dataset [20] includes 5,094 short-exposure images in RAW format, each matched with a corresponding long-exposure reference. The Ex Dark dataset [22] comprises 7,363 unlabeled low-light images categorized into 12 object classes, adapted for tasks involving low-light detection and recognition. The UHD-LOL dataset [24] includes two subsets, UHDLOL4K and UHD-LOL8K, corresponding to resolutions of 4K and 8K, respectively. The PNIL dataset [23] contains 2000 paired images, offering higher resolution and more diverse scenes compared to previous datasets.

Table 2 Datasets used for LLIE

Year	Dataset	Quantity	Resolution	Format	Paired
2018	SID[20]	5,094	4,240×2,832/6000×4000	RAW	Yes
2018	LOL[21]	789	400×600	RGB	Yes
2019	ExDark[22]	7,363	500×332	RGB	No
2022	PNIL[23]	2,000	6,720×4,480	RGB	Yes

Year	Dataset	Quantity	Resolution	Format	Paired
2023	UHD- LOL[24]	11,065	4,000×4,000/8,000×8,000	RGB	Yes

IV. EVALUATION METRICS

Image enhancement performance is typically evaluated through both subjective and objective metrics. Subjective assessment relies on human visual perception to judge enhancement effectiveness, while objective evaluation employs quantitative measurements. Participants need to evaluate the quality, brightness, clarity, and other aspects of the image, which is intuitive and highly credible. However, due to individual and environmental influences, the results have a certain degree of uncertainty.

Objective evaluation is based on computer vision and image processing to assess the effectiveness of image enhancement algorithms.

Mean Square Error (MSE) [24]: Used to measure the degree of difference between the original image and the enhanced image, i.e., the mean squared error. The smaller the value, the better the enhancement effect. The formula is:

$$MSE = \frac{1}{nm} \sum_{i=1}^m \sum_{j=1}^n (I(i,j) - K(i,j))^2$$

Where m and n represent the image dimensions in rows and columns, where $I(i,j)$ and $K(i,j)$ denote pixel values at position (i,j) in the original and enhanced images, respectively.

Peak Signal-to-Noise Ratio (PSNR) [24]: Quantifies the noise ratio between original and enhanced images, where higher values correspond to superior enhancement quality. The formula is:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

Among them, MAX_I is the maximum possible image pixel value.

Structural Similarity Index Measure (SSIM) [25]: It provides a numerical score reflecting how structurally similar the enhanced image is to the original; a score nearing 1 signifies the best possible enhancement performance in terms of structure. The formula is:

$$SSIM(x,y) = \left(\frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \right) \cdot \left(\frac{2\sigma_{xy} + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \right)$$

For the comparison between two images, X and Y , μ_x and μ_y are the average brightnesses, μ_x^2 and μ_y^2 are the variances, μ_{xy} are the covariances, and C_1 and C_2 are small constants added to avoid zero denominator.

Learned Perceptual Image Patch Similarity (LPIPS) [26]: A deep learning-based metric for image quality assessment that evaluates the perceptual similarity between two images to quantify their visual differences. LPIPS is closer to human visual perception and can effectively capture the subtle differences between images, but its limitation is it rely on pre-trained model and has high computation.

Natural Image Quality Evaluator (NIQE) [27]: A statistical model-based objective evaluation metric that can

automatically assess image quality, with smaller values indicating better enhancement effects and higher consistency with human subjective evaluation results. The formula is:

$$NIQE = \sqrt{(v - \mu_{train})^T \Sigma_{train}^{-1} (v - \mu_{train})}$$

Where v : The feature vector of the test image (36-dimensional, obtained from the statistics of local normalization coefficients).

μ_{train} : The mean vector of the features from the natural image training set.

$\Sigma_{\{train\}}$: The covariance matrix of the natural image training set.

Table 3 Quantitative comparison of methods

Method	SSIM↑	PSNR↑	LPIPS↓	NIQE↓
Zero-DCE[6]	0.9035	19.8183	0.335	5.596
DSLR[7]	0.9034	19.7315	0.318	4.23
DBRN[8]	0.474	15.125	0.316	5.121
RUAS [9]	0.8925	19.6934	0.362	5.625
SCI [10]	0.8836	19.4468	0.298	3.308
SNR-aware [11]	0.831	23.17	0.225	4.02
LLFormer [12]	0.8163	23.6491	0.1692	3.89
PairLIE [13]	0.736	19.51	0.248	3.78
CUE[14]	0.67	20.38	0.245	3.95
Dimma[15]	0.85	23.41	0.22	3.92
CDAN[16]	0.818	20.102	0.167	4.12
Cycle-Retinex[17]	0.9438	20.7572	0.285	4.15

In Table 3, we provide a quantitative comparison of the different methods introduced here, using the LOL-test dataset, the results come from the original papers of these methods. The best performance scores within the table are marked with bold formatting.

LLFormer performs best in PSNR, indicating its significant advantage in maintaining image details and restoration quality, while also showing good performance in LPIPS and NIQE; CDAN outperforms in perceptual loss (LPIPS), meaning the images it generates are visually more natural; Cycle-Retinex has the highest SSIM, indicating the best structural preservation; SCI performs best in NIQE, suggesting the highest natural image quality, with a

balanced overall performance; DBRN performs poorly across all metrics, especially in PSNR and LPIPS, indicating its inadequacy in low-light enhancement.

As presented in Table 3, methods based on supervised learning demonstrate superior performance in full-reference metrics (SSIM, PSNR, LPIPS and NIQE). No single algorithm achieves the best performance across all four-evaluation metrics. This result indicates that different enhancement methods have varying focuses, and it is rare to find a method that outperforms in all aspects.

V. CONCLUSION

This paper presents a comprehensive and systematic review of the LLIE field from four key aspects: LLIE methods, benchmark datasets, evaluation metrics, and quantitative comparison results. Despite significant progress, several practical limitations still hinder the deployment of LLIE techniques in real-world applications. Current deep learning-based LLIE models often require extensive computational resources and large-scale annotated datasets, limiting their adaptability to resource-constrained environments such as mobile or embedded platforms. Moreover, these models may not generalize well to diverse lighting conditions, domains, or downstream vision tasks.

To address these challenges and advance the field, future research can focus on the following directions:

1. Enhancing the robustness and adaptability of LLIE algorithms to handle unseen environments and varying illumination conditions.
2. Developing lightweight and efficient models with fewer parameters to enable real-time processing on low-power devices.
3. Building more diverse and task-specific datasets through synthetic augmentation and realistic simulation, to better reflect real-world complexity.
4. Establishing tighter integration between LLIE and high-level vision tasks (e.g., detection, segmentation) to form unified, end-to-end pipelines that are jointly optimized.

By overcoming these limitations, LLIE systems can become more practical, scalable, and impactful in real-world computer vision applications.

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